

EKIMCHAN, Russian Federation

WMO: 313290

Lat: **53.0749N**Lon: **132.9510E**Elev: **1779**StdP: **13.78**Time Zone: **9.00 (E09)**Period: **94-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-42.9	-39.4	-47.3	0.4	-42.3	-43.7	0.5	-39.0	12.7	6.9	9.3	-5.0	1.1	70	0.651

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	21.1	83.6	64.3	80.1	63.3	76.6	61.8	67.9	76.8	66.2	74.7	64.6	72.7	5.2	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
65.3	99.9	70.4	63.6	93.9	69.1	61.8	88.0	67.9	33.4	76.7	32.1	74.6	30.7	72.6	74.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 13.9	11.5	9.6	-47.1	87.9	3.8	2.9	-49.8	90.1	-52.0	91.8	-54.1	93.4	-56.9	95.6		
(5)			-46.7	70.5	3.7	2.1	-49.3	72.0	-51.4	73.2	-53.5	74.4	-56.2	75.9		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	25.0	-20.0	-8.9	8.6	29.2	46.6	58.7	65.1	60.8	47.4	28.4	0.9	-19.5
	DBStd	31.72	10.44	9.63	10.48	9.04	7.41	6.38	5.23	5.74	7.67	9.64	13.71	11.32
	HDD50	10329	2171	1648	1284	623	155	9	0	3	137	671	1473	2155
	HDD65	14725	2636	2068	1749	1073	571	204	66	152	527	1136	1923	2620
	CDD50	1187	0	0	0	1	49	271	467	339	60	0	0	0
	CDD65	107	0	0	0	0	1	16	68	22	0	0	0	0
	CDH74	1291	0	0	0	1	40	307	711	228	4	0	0	0
	CDH80	295	0	0	0	0	4	64	195	32	0	0	0	0
	WSAvg	3.1	1.5	2.3	3.5	4.3	4.5	3.6	3.1	3.0	3.3	3.5	2.5	1.5
	PrecAvg	27.3	0.3	0.3	0.5	1.5	2.7	3.7	5.7	5.5	3.5	1.7	0.9	0.5
(15) Precipitation	PrecMax	36.5	1.0	1.9	1.1	3.2	5.1	9.1	13.8	10.8	7.4	5.1	2.0	1.1
	PrecMin	22.3	0.0	0.0	0.1	0.1	0.9	0.9	1.1	1.3	0.7	0.3	0.3	0.0
	PrecStd	3.2	0.2	0.3	0.2	0.7	1.1	1.8	2.4	2.3	1.5	0.9	0.4	0.2
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	17.6	23.2	41.4	63.9	79.6	85.7	89.2	83.7	73.4	59.5	32.6	20.5
		MCWB	15.1	20.1	32.5	45.9	56.3	64.6	66.7	66.0	59.1	46.7	29.9	18.2
	2%	DB	5.4	17.7	35.3	55.5	73.0	81.6	85.2	79.7	68.5	50.7	28.7	9.0
		MCWB	4.1	15.1	28.6	42.4	53.1	62.0	65.3	64.3	56.9	41.6	26.8	7.8
	5%	DB	-0.3	12.5	30.9	49.2	68.2	77.4	81.6	76.2	64.3	45.0	25.2	2.4
		MCWB	-1.2	10.2	25.7	38.1	50.9	59.3	64.7	63.1	54.7	38.9	23.6	1.5
	10%	DB	-5.0	7.4	26.9	44.2	63.1	73.4	77.9	72.5	60.1	40.8	21.1	-2.5
		MCWB	-5.7	5.6	22.6	35.8	48.3	58.6	64.0	61.9	52.5	36.0	19.7	-3.2
	0.4%	WB	14.9	20.2	33.0	47.8	57.7	68.5	70.9	69.4	61.8	47.7	31.1	18.3
		MCDB	17.8	23.2	39.9	60.2	74.4	79.0	80.2	78.9	68.9	56.4	32.6	20.3
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	4.3	15.2	29.2	43.3	55.0	65.1	68.4	67.2	58.6	43.3	27.0	7.9
		MCDB	5.4	17.8	34.5	54.3	69.1	75.9	77.6	74.9	65.5	48.7	28.6	9.0
	5%	WB	-1.1	10.3	26.0	39.1	52.4	62.8	66.9	65.4	56.5	39.7	23.5	1.6
		MCDB	-0.3	12.5	30.8	47.7	64.0	71.7	75.7	72.0	62.2	44.2	25.1	2.4
	10%	WB	-5.6	5.7	22.7	36.1	50.0	60.4	65.5	63.6	53.9	36.5	19.7	-3.2
		MCDB	-5.0	7.4	26.8	43.6	60.2	69.0	74.1	69.7	58.6	40.2	21.1	-2.5
(35) Mean Daily Temperature Range	5% DB	MDBR	19.5	26.7	27.1	21.8	23.3	23.5	21.1	20.0	19.4	16.9	17.1	16.0
		MCDBR	19.5	27.0	28.7	28.9	34.1	31.2	28.7	26.5	25.5	24.5	15.3	14.6
		MCWBR	18.2	24.0	21.9	16.4	16.2	12.7	10.8	11.8	13.4	14.9	13.1	13.7
		MCDBR	19.2	26.4	26.7	26.9	29.0	24.0	22.1	20.5	21.6	22.3	14.8	14.5
	5% WB	MCWBR	18.0	23.4	20.5	15.7	14.7	11.4	10.5	10.7	12.4	14.2	13.0	13.6
(40) Clear-Sky Solar Irradiance	taub		0.249	0.285	0.347	0.474	0.526	0.489	0.475	0.412	0.373	0.376	0.299	0.249
	taud		2.125	2.062	1.923	1.730	1.821	2.018	2.106	2.294	2.324	2.081	2.037	2.066
	Ebn at Noon		231	255	258	236	232	244	245	257	252	213	197	201
	Edh at Noon		22	32	46	65	62	52	47	36	31	32	24	19
(44) All-Sky Solar Radiation	RadAvg		428	799	1325	1665	1578	1755	1599	1339	990	622	426	322
	RadStd		11	23	52	119	162	173	137	127	103	62	35	11

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (4 neighbors)	N/A	N/A	N/A	N/A	+0.85	+1.10	N/A	N/A	N/A	N/A

Nomenclature: See separate page